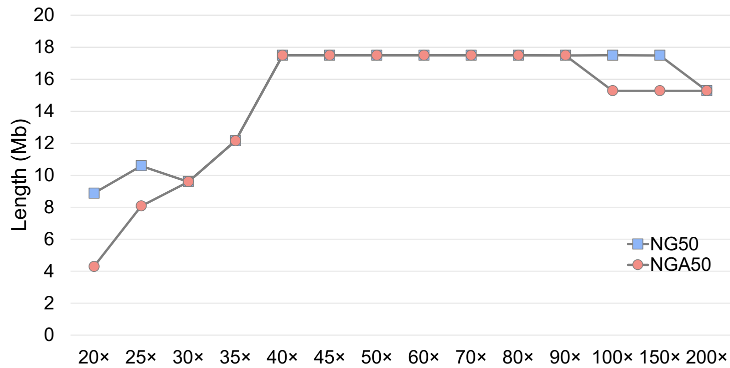
Towards optimal selection of ultra-deep sequencing reads for *de novo* genome assembly

(Supplemental Material)



Supplemental Figure 1: NG50 and NGA50 of the assemblies produced by running hifiasm on various subset of reads selected by AWin$K$ at different coverage depths from a set of PBSim simulated HiFi data with sequencing depth of 1,000× for the genome of *C. elegans*

Supplemental Table 1: The length of the six longest contigs, number of assembled contigs and the total assembly length produced by hifiasm on a subset of reads selected (i) at random (200× coverage), (ii) with best average QV (200×), (iii) with longest (200×), and (iv) produced by the "oracle" at 5×, 10×, and 20× coverage selected from a set of PBSim synthetic HiFi reads with a sequencing depth of 1000× for the genome of *C. elegans* (contig1 is chromosome V)

Assembly	contig1	contig2	contig3	contig4	contig5	contig6	# of contigs	Total length (bp)
Reference	20,924,180	17,718,942	17,493,829	15,279,421	15,072,434	13,783,801	7	100,286,401
Random (200×)	18,416,632	17,718,798	17493,796	15279214	15072421	13,783,719	9	100,285,249
QV: best (200×)	18,379,014	17,718,881	17,493,792	15,279,304	15,072,395	13,783,687	9	100,239,133
Longest (200×)	18,389,640	17,718,888	17,493,256	15,279,172	15,072,425	13,783,562	8	100,216,292
oracle (5×)	35,032,850	23,002,036	15,273,332	13,406,979	3,983,934	1,667,497	8	92,427,632
oracle (10×)	20,815,879	17,718,653	17,493,341	15,279,162	15,072,238	13,775,179	7	100,188,613
oracle (20×)	20,816,019	17,718,829	17,493,548	15,279,162	15,072,238	13,783,620	7	100,198,975

Supplemental Table 2: Length of the six longest assembled contigs, number of assembled contigs and the total assembly length produced by hifiasm on subset of reads selected by AWin$K$ at different sequencing depths from synthetic HiFi reads with sequencing depth of 1000× for the genome of *C. elegans*

Assembly	contig1	contig2	contig3	contig4	contig5	contig6	# of contigs	Total length (bp)
Reference	20,924,180	17,718,942	17,493,829	15,279,421	15,072,434	13,783,801	7	100,286,401
20×	14,353,301	10,245,828	9,636,687	9,618,046	8,864,646	8,439,682	64	94,317,956
25×	14,842,596	14,353,669	12,142845	10,579,969	8,345,395	8,066,091	39	100,062,442
30×	17,492,666	14,353,670	10,983157	9,582,624	9,535,291	8,345,394	25	100,130,912
35×	17,718,886	17,492,967	13,232487	12,142,865	11,008,444	10,522,223	19	100,166,017
40×	18,389,194	17,718,886	17,492965	13,232,438	12,154,690	11,009,585	18	100,244,671
45×	18,389,193	17,718,886	17,493058	15,071,354	11,977,560	11,113,608	18	100,435,205
50×	18,460,696	17,718,888	17,493086	15,276,212	15,071,363	11,977,763	16	100,481,717
60×	18,389,436	17,701,132	17,493086	15,276,206	15,071,359	13,781,138	37	101,063,716
70×	18,389,436	17,710,003	17,493057	15,276,213	15,071,359	13,783,520	93	102,553,110
80×	18,389,641	17,718,525	17,493057	15,252,239	15,071,357	13,783,727	109	102,668,652
90×	18,389,641	17,708,807	17,490415	15,276,211	15,071,357	13,783,727	98	102,153,158
100×	20,774,037	17,707,649	17,493695	15,276,312	15,071,358	13,783,791	189	104,474,733
150×	19,800,194	17,711,149	17,488195	15,276,393	15,062,976	13,783,793	414	109,417,601
200×	17,718,525	17,488,195	15,276394	15,063,184	13,783,791	11,299,277	641	115,281,081

Supplemental Table 3: Comparing the NG50 and NGA50 metrics for hifiasm assemblies produced by various read selection strategies for *C. elegans*

Method	**Depth**	**Performance metrics**	
		NG50	NGA50
SLICES	50×	17,493,698	17,493,698
	100×	17,493,661	17,493,649
RANDOM	50×	17,493,493	17,493,493
	100×	17,493,731	15,279,061
QV	50×	17,492,865	17,492,865
	100×	17,493,621	17,493,621
YAK	50×	17,493,202	15,274,813
	100×	17,493,202	15,276,768
Longest	50×	17,492,955	17,492,955
	100×	17,492,985	17,492,985
Filtlong	50×	17,492,983	17,492,983
	100×	17,493,188	17,493,188
AWin$K$	50×	17,493,086	17,493,086
	100×	17,493,695	15,276,312

Supplemental Table 4: Average length of short contigs produced by hifiasm on subset of reads selected by AWin$K$ at sequencing depths = $\{100\times, 150\times, 200\times\}$ from synthetic HiFi reads with sequencing depth of 1000× for the genome of *C. elegans*

Depth	**# of contigs**	**Length < 50 kb**		**Length < 100 kb**	
		# of contigs (%)	avg. length (bp)	# of contigs (%)	avg. length (bp)
100×	189	175 (92%)	21,730.80	183 (97%)	23,868.30
150×	414	390 (94%)	21,213.00	407 (98%)	22,919.10
200×	641	601 (94%)	21,588.30	630 (98%)	23,516.70